

Tutorial 11 – Program Design

1. Name the SEVEN levels of cohesion in program design. Which of the levels mentioned is most desirable and least desirable?
2. List and briefly explain any 6 types of cohesion. You may substantiate your explanation with a practical example.
3. Define the following terms.
 - i. Pathological coupling
 - ii. Simple data coupling
 - iii. External coupling
 - iv. Communicational coupling
 - v. Functional cohesion
 - vi. Coincidental cohesion
 - vii. Logical cohesion
 - viii. Temporal cohesion
4. What is coupling? List and describe any 3 levels of coupling.
5. Define Tight Coupling and Loose Coupling. Which is the most intimate way to share data?
6.
 - i. Describe any 2 levels of cohesion.
 - ii. What is the highest level of cohesion?
 - iii. What is the lowest level of cohesion?
7. When deciding how to organize your program into modules, you should follow two general rules, namely reduce coupling and increase cohesion. Explain the meanings of the terms coupling and cohesion.
8. Write down the level of coupling or cohesion for each of the following descriptions:
 - i. This level of coupling is the loosest type. It is also known as normal coupling. It occurs when modules share a data item by passing parameters
 - ii. This level of coupling occurs when a main program (or other module) passes a parameter to a module, controlling the module's actions or telling it what to do.

- iii. This level of coupling occurs when two or more modules change one another's data. It is the tightest coupling among all.
 - iv. This level of cohesion occurs when all operations in a module contribute to the performance of only one task. It is the highest level of cohesion among all.
 - v. This level of cohesion takes place when a member module performs one or more tasks depending on a decision, where the decision is in the form of a case structure or a series of if statements.
9. The 7 types of cohesion in alphabetical order are: coincidental cohesion, communicational cohesion, functional cohesion, logical cohesion, procedural cohesion, sequential cohesion, and temporal cohesion. Identify the correct cohesion type for parts (i) to (vii) in the table below.

Characteristics			Type of cohesion
The module performs a single task.			(i)
The module does not perform a single task.	Data relates the activities inside the module.	Sequence is important.	(ii)
		Sequence is not important.	(iii)
	Logic control relates the activities inside the module.	Sequence is important.	(iv)
		Sequence is not important.	(v)
	Neither data nor logic relates the activities inside the module.	Activities are related by time.	(vi)
		Activities are not related by time.	(vii)